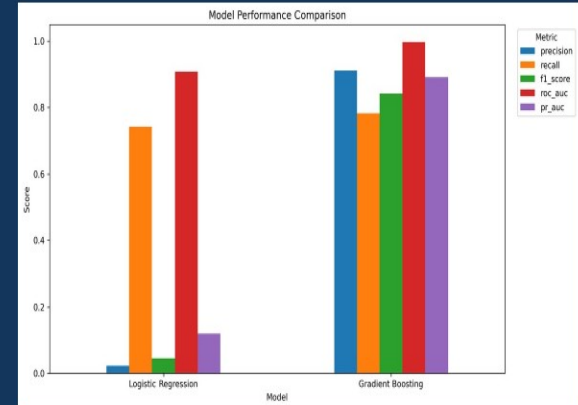


FraudShield BI

Quantifying the Impact of False Positives in Credit Card Fraud Detection

MIS581 Capstone | Business Intelligence and Data Analytics

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Colorado State University Global
Dr. Jamia Mills | July 2026



91.14%

Gradient boosting
precision at 0.50

163

False positives at 0.50

False positives create a hidden “tax” through review workload, customer friction, and missed business opportunities.

Business Problem: Fraud Controls Can Create Customer Friction

The practical challenge

Fraud systems must identify suspicious activity quickly.

Overly aggressive controls can block or review legitimate customers.

False positives can increase manual reviews, call volume, customer frustration, and lost transaction value.

Accuracy alone can hide these tradeoffs in highly imbalanced fraud data.



FraudShield BI reframes fraud detection as a business decision:

Model choice + threshold choice → fraud prevention, customer friction, review workload, and cost.

False Positive Tax

A business-readable estimate of the cost created when legitimate transactions are incorrectly flagged.

Research Questions and Hypotheses

RQ1

How does changing the fraud threshold affect false positives, fraud detection, and estimated False Positive Tax?

H01 rejected**RQ2**

Which transaction characteristics and model outputs are associated with false-positive decisions?

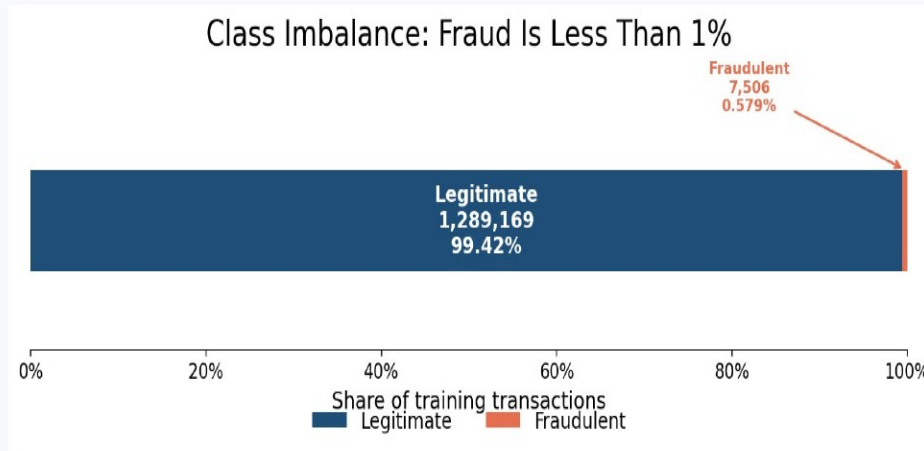
H02 rejected**RQ3**

How can BI dashboards and scenario analysis support fraud-prevention decisions?

H03 rejected

Analytical approach: confusion matrices, threshold analysis, precision-recall metrics, chi-square tests, Mann-Whitney U tests, cost-scenario analysis, and Power BI reporting.

Dataset, Research Design, and Ethical Boundaries

**1,296,675**

training records

555,719

testing records

0.579%

training fraud rate

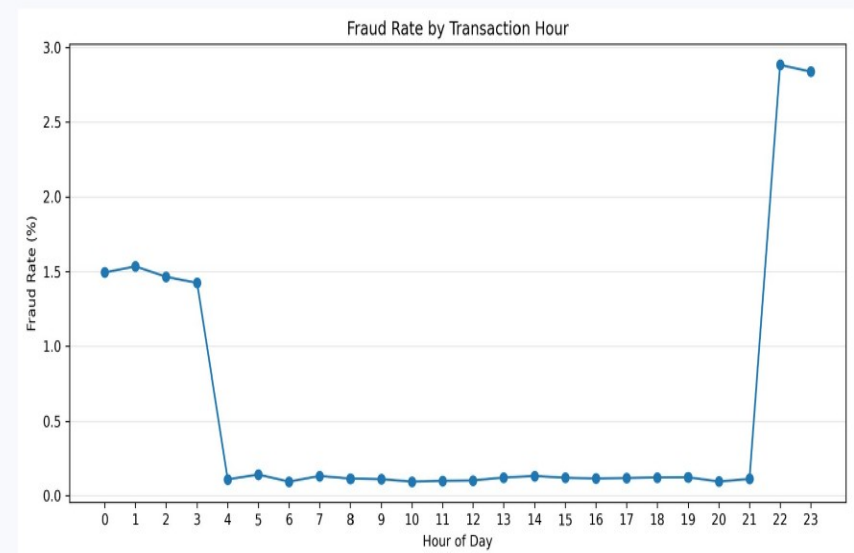
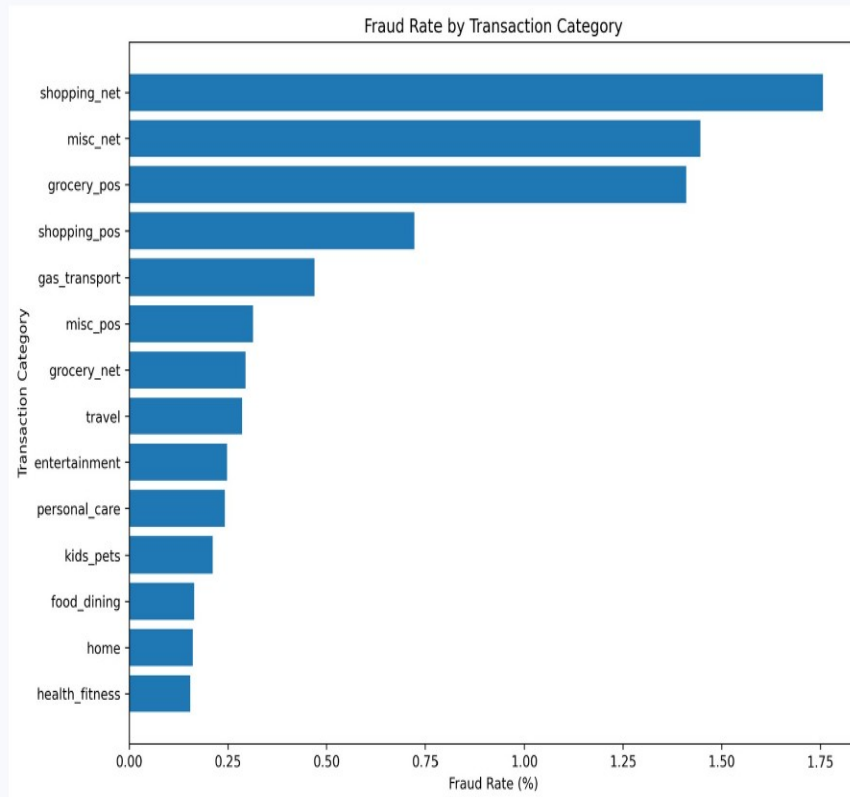
Design choices

Quantitative secondary analysis using a simulated Kaggle/Sparkov transaction dataset.
Pre-split train/test files used; production work should add validation and holdout samples.
Predictors included amount, hour, age, city population, distance, category, and gender.

Ethical guardrails

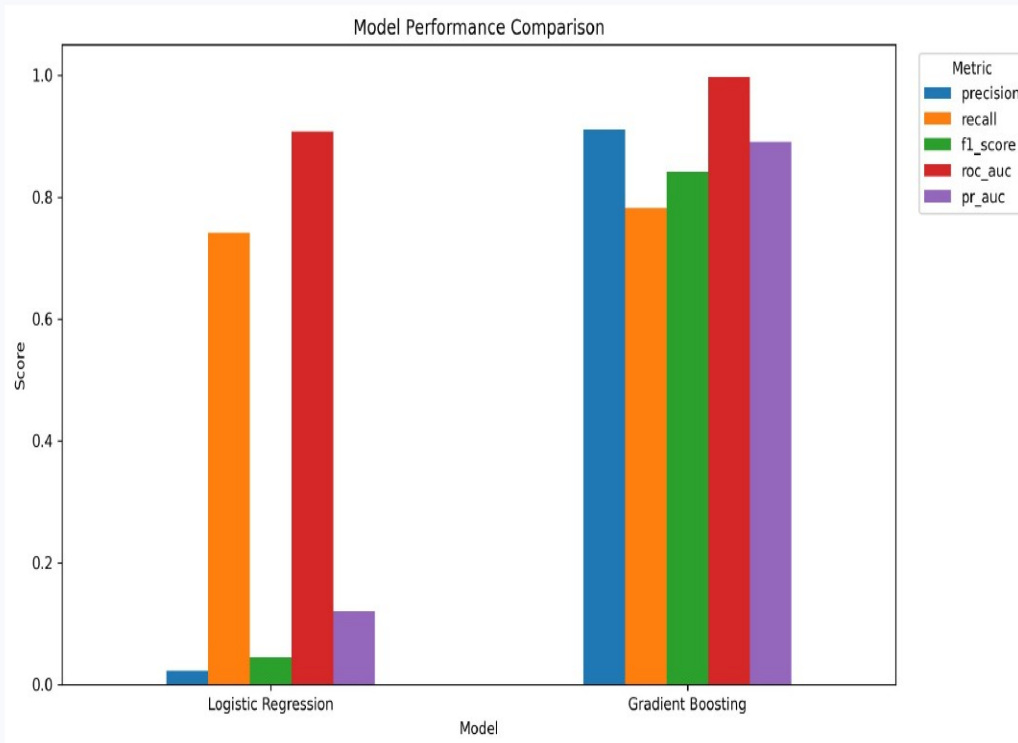
No employer records, customer accounts, or internal fraud rules were used.
Direct identifiers and high-cardinality fields were excluded from modeling.
Dashboard is positioned as decision support, not fully automated production approval.

Exploratory Findings: Fraud Patterns Were Segment-Specific



Key pattern: online shopping, miscellaneous online, and grocery POS had the highest fraud rates. Fraud also rose sharply during late-night hours.

Model Comparison: Gradient Boosting Was the Clear Winner



Default 0.50 threshold

67,204

logistic false positives

163

gradient boosting false positives

2.31%

logistic precision

91.14%

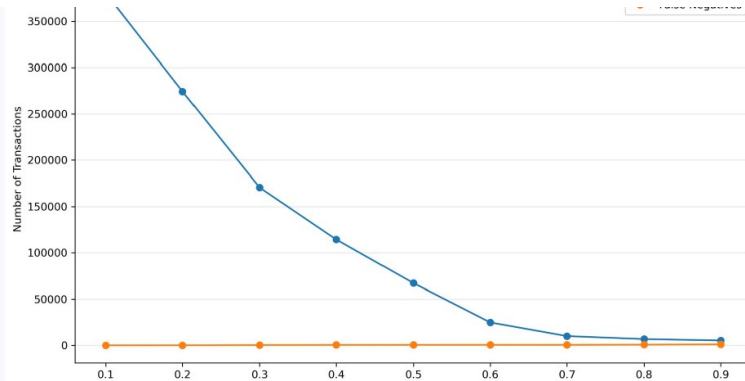
gradient boosting precision

At the same threshold, gradient boosting detected more fraud and reduced false positives by 67,041 compared with logistic regression.

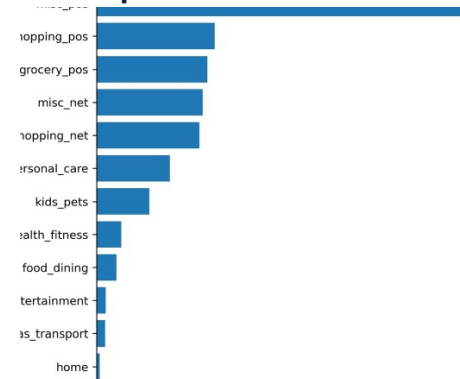
Threshold and False-Positive Impact

The cutoff changed review workload, fraud capture, and customer friction.

Threshold error tradeoff



False positives were concentrated



13.02%

travel false-positive rate

2.83%

highest hourly FP rate

\$640.94

mean FP amount

Operational takeaway

Thresholds should reflect cost, friction, and review capacity.
High-impact segments may need step-up authentication instead of automatic decline.

Lower thresholds catch more fraud but can increase false positives; higher thresholds reduce alerts but may miss fraud.

Cost Scenario Analysis and Power BI Decision Support

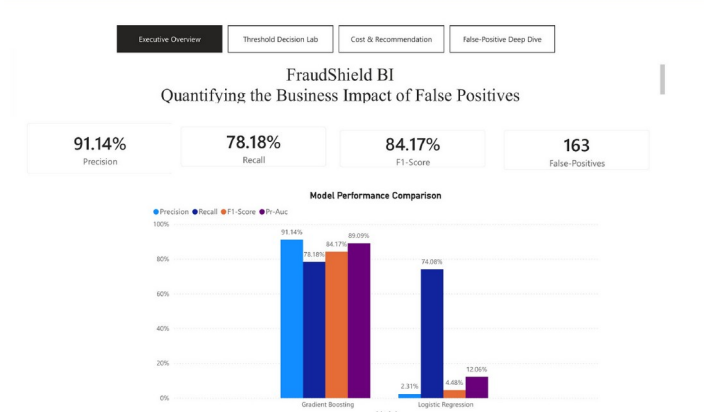
Gradient boosting at a lower threshold minimized estimated cost while keeping the tradeoffs visible to business stakeholders.

Moderate-cost recommendation



Recommended: Gradient Boosting at 0.10 threshold

Power BI decision support



What the dashboard adds

Executive view of model KPIs • threshold lab for review volume and cost • false-positive deep dive by category and hour

Dashboard values were validated against Python exports before final reporting.

Conclusion, Validation Notes, and Final Package

What the project showed

Accuracy alone is misleading with fraud below 1%.
Gradient boosting outperformed logistic regression.
Threshold choice changed fraud capture, workload, and cost.
Power BI made tradeoffs clear for business stakeholders.

Future validation notes

Use validation and untouched holdout samples.
Expand FP cost beyond fixed per-alert amounts.
Repeat segmentation using the selected model.
Evaluate fairness with and without sensitive attributes.

GitHub-ready package

README and run instructions.
Notebook, validated outputs, and charts.
Final paper and presentation materials.
Power BI PBIX, PDF export, and page previews.

Final recommendation: govern model and threshold choices with fraud loss, customer friction, operational capacity, and fairness in view.

GitHub repository: <https://github.com/kaymerc/fraudshield-bi-false-positive-tax>

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Additional technical documentation used: Microsoft Power BI and scikit-learn documentation.